Mathematical Analysis — Midterm 2025, Row 2

Past exam · Răzvan

> **How to use.** The statement is on this page, exactly as in the official PDF (Class Materials → Past Exams → *midterm_2025_2.pdf*). Solve it on paper in **1h50min**, then turn to the solutions, which come after a red STOP page.

Midterm Test 2025 — Row 2

1. Let $A \subseteq \mathbb{R}$ be a bounded set. Prove that $\forall \varepsilon > 0$, $\exists x \in A$ such that $x < \inf(A) + \varepsilon$. 1p

2. Study the convergence of the following series: (a) $\displaystyle\sum_{n=1}^{\infty} \frac{a^n}{a + n^2}$, $a > 0$. (b) $\displaystyle\sum_{n=1}^{\infty} \frac{2 \cdot 4 \cdot \ldots \cdot (2n)}{1 \cdot 3 \cdot \ldots \cdot (2n-1)}$. 4p
   (c) $\displaystyle\sum_{n=1}^{\infty} n \left(\sin \frac{1}{n} - \sin \frac{1}{n+1} \right)$. (d) $\displaystyle\sum_{n=2}^{\infty} \frac{\ln n^2}{n^2}$.

3. Compute $\displaystyle\int_0^{\infty} x^3 e^{-x^2}\, dx$. 1p

4. (a) Find the Taylor series around 0 for $\cos^2 x$. (1p)
   (b) Find the higher order derivatives $f^{(n)}(0)$, $n \in \mathbb{N}$, for the function (0.5p)

$$f(x) = \begin{cases} \dfrac{\sin x}{x}, & x \neq 0 \\ 1, & x = 0. \end{cases}$$

5. Find the radius of convergence and the sum of the power series $\displaystyle\sum_{n=1}^{\infty} (-1)^n n^2 x^n$. 1.5p

Time 1h50min. Marking starts from 1p. Total 10p.

⛔ STOP
SOLUTIONS BELOW

De aici încep rezolvările

Rezolvă mai întâi subiectul singur, cronometrat (1h50min), și abia apoi dă pagina. Toate paginile cu rezolvări au o bandă roșie în stânga.

SOLUTIONS — Midterm 2025, Row 2

1. Approximation property of the infimum

A is non-empty and bounded, so $m = \inf A$ exists; m is the *greatest* lower bound of A.

Let $\varepsilon > 0$ and suppose, by contradiction, that $x \geq m + \varepsilon$ for every $x \in A$. Then $m + \varepsilon$ is a lower bound of A and $m + \varepsilon > m$ — contradicting that m is the greatest lower bound.

$\Longrightarrow \exists x \in A$ **with** $x < \inf A + \varepsilon$**.** ■

2. Convergence of series

(a) Ratio test:

$$\frac{a_{n+1}}{a_n} = a \cdot \frac{a + n^2}{a + (n+1)^2} \to a.$$

- $a < 1$: converges; $a > 1$: diverges.
- $a = 1$: the series is $\sum \frac{1}{1+n^2}$, and $\frac{1}{1+n^2} < \frac{1}{n^2}$, so it converges (comparison).

$\Longrightarrow$ **converges** $\iff 0 < a \leq 1$**. (Compare with Row 1, where** n **instead of** n^2 **made** $a = 1$ **divergent.)**

(b) Every factor $\frac{2k}{2k-1} > 1$, so $a_n = \prod_{k=1}^{n} \frac{2k}{2k-1} > 1$ for all n. Hence $a_n \not\to 0$.

$\Longrightarrow$ **diverges (necessary condition fails). No ratio or Raabe test is needed.**

(c) By the Mean Value Theorem, as in Row 1:

$$\sin\frac{1}{n} - \sin\frac{1}{n+1} = \frac{\cos\xi_n}{n(n+1)}, \quad \xi_n \in \left(\tfrac{1}{n+1}, \tfrac{1}{n}\right) \implies a_n = \frac{\cos\xi_n}{n+1} > 0.$$

Limit comparison with $\frac{1}{n}$: $\frac{a_n}{1/n} = \frac{n\cos\xi_n}{n+1} \to 1$, and $\sum \frac{1}{n}$ diverges.

$\Longrightarrow$ **diverges.**

(d) $a_n = \frac{\ln n^2}{n^2} = \frac{2\ln n}{n^2} > 0$. Since $\frac{\ln n}{\sqrt{n}} \to 0$, there is N with $\ln n \leq \sqrt{n}$ for $n \geq N$ (in fact for all $n \geq 1$). Then

$$0 < a_n \leq \frac{2\sqrt{n}}{n^2} = \frac{2}{n^{3/2}}, \qquad \sum \frac{1}{n^{3/2}} \text{ converges.}$$

$\Longrightarrow$ **converges (comparison).**

Rule of thumb: logarithms grow slower than any power, so $\sum \frac{(\ln n)^q}{n^p}$ converges for every q whenever $p > 1$.

3. Improper integral

Substitute $t = x^2$, $dt = 2x\,dx$, so $x^3\,dx = x^2 \cdot x\,dx = \frac{t}{2}\,dt$; $t : 0 \to \infty$:

$$\int_0^\infty x^3 e^{-x^2}\,dx = \frac{1}{2}\int_0^\infty te^{-t}\,dt = \frac{1}{2}\lim_{b\to\infty}\Big[-(t+1)e^{-t}\Big]_0^b = \frac{1}{2}.$$

$\Longrightarrow$ **the integral converges and equals** $\frac{1}{2}$**.**

4. Taylor series and derivatives at 0

(a) $\cos^2 x = \frac{1+\cos 2x}{2}$, with $\cos 2x = \sum_{n\geq 0}(-1)^n \frac{2^{2n}x^{2n}}{(2n)!}$:

$$\cos^2 x = 1 + \sum_{n=1}^{\infty}(-1)^n \frac{2^{2n-1}}{(2n)!}x^{2n} = 1 - x^2 + \frac{x^4}{3} - \frac{2x^6}{45} + \dots$$

$\Longrightarrow$ valid for all $x \in \mathbb{R}$ ($R = \infty$). Check: $\sin^2 x + \cos^2 x = 1$ term by term (compare with Row 1).

(b) For $x \neq 0$:

$$\frac{\sin x}{x} = \frac{1}{x}\sum_{k\geq 0}(-1)^k \frac{x^{2k+1}}{(2k+1)!} = \sum_{k\geq 0}(-1)^k \frac{x^{2k}}{(2k+1)!} = 1 - \frac{x^2}{6} + \frac{x^4}{120} - \dots$$

At $x = 0$ the series gives $1 = f(0)$. So f is the sum of a power series with $R = \infty$, hence C^∞, and $f^{(m)}(0) = m!\, c_m$:

$$f^{(2k+1)}(0) = 0, \qquad f^{(2k)}(0) = (2k)! \cdot \frac{(-1)^k}{(2k+1)!} = \frac{(-1)^k}{2k+1}, \quad k \in \mathbb{N}.$$

$\Longrightarrow$ $f(0) = 1$, $f''(0) = -\frac{1}{3}$, $f^{(4)}(0) = \frac{1}{5}$, ...; odd derivatives are 0.

5. Power series $\sum(-1)^n n^2 x^n$

Radius: $|c_n| = n^2$, $\sqrt[n]{n^2} \to 1$ $\Longrightarrow$ $R = 1$; at $x = \pm 1$ the terms have absolute value $n^2 \not\to 0$ $\Longrightarrow$ diverges. Convergence set $(-1, 1)$.

Sum. $(-1)^n n^2 x^n = n^2(-x)^n$, so with $y = -x$ ($|y| < 1$) and the formula $\sum n^2 y^n = \frac{y(1+y)}{(1-y)^3}$ (derived by differentiating the geometric series twice, see Row 1):

$$\sum_{n=1}^{\infty}(-1)^n n^2 x^n = \frac{-x(1-x)}{(1+x)^3} = \frac{x(x-1)}{(1+x)^3}, \qquad |x| < 1.$$

$\Longrightarrow$ $R = 1$, sum $\frac{x(x-1)}{(1+x)^3}$. (Check: $x = 0.3$ gives -0.0956 both ways.)

Results at a glance

Ex.	Answer	Tool
1	else $\inf A + \varepsilon$ would be a larger lower bound	definition of inf
2	(a) iff $a \leq 1$; (b) diverges; (c) diverges; (d) converges	ratio; $a_n > 1$; MVT + comparison with $\frac{1}{n}$; $\ln n \leq \sqrt{n}$
3	$\frac{1}{2}$	$t = x^2$, by parts
4	$\cos^2 x = 1 + \sum(-1)^n \frac{2^{2n-1}x^{2n}}{(2n)!}$; $f^{(2k)}(0) = \frac{(-1)^k}{2k+1}$, odd $= 0$	linearization; coefficients
5	$R = 1$, $\frac{x(x-1)}{(1+x)^3}$	substitute $y = -x$